

Math 520, F07
Quiz 1

Name: Answer
SSN: xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. Please simplify your answer if possible.

Problem 1. (10 points)

a) Find the general solution of the given differential equation:

$$\frac{dy}{dt} + yt \sin t^2 = 0.$$

Solution:

$$y(t) = C e^{-\int t \sin t^2 dt}$$
$$= C e^{\frac{1}{2} \cos t^2}$$

b) Find the specific solution of the given initial-value problem:

$$\frac{dy}{dt} + yt \sin t^2 = 0, \quad y(1) = 2.$$

Solution: Method 1: Put $y(1) = 2$ into the general solution obtained in a), we have

$$y(1) = C e^{\frac{1}{2} \cos 1} = 2$$
$$\Rightarrow C = 2 e^{-\frac{1}{2} \cos 1}$$

$$\text{So } y(t) = 2 e^{-\frac{1}{2} \cos 1} \cdot e^{\frac{1}{2} \cos t^2} = 2 e^{\frac{1}{2} (\cos t^2 - \cos 1)}$$

Method 2:

$$y(t) = y(1) e^{-\int_1^t s \sin s^2 ds}$$
$$= 2 e^{\frac{1}{2} \cos s^2 \Big|_{s=1}^{s=t}}$$
$$= 2 e^{\frac{1}{2} (\cos t^2 - \cos 1)}$$

Problem 2. (10 points)

a) Find the general solution of the given differential equation:

$$\frac{dy}{dt} - 2ty = t.$$

Solution: $\mu(t) = e^{\int -2t dt} = e^{-t^2} = e^{-t^2}$

$$\begin{aligned}\Rightarrow y(t) &= \frac{1}{\mu(t)} \left(\int \mu(t) b(t) dt + C \right) \\ &= e^{t^2} \left(\int e^{-t^2} t dt + C \right) \\ &= e^{t^2} \left(-\frac{e^{-t^2}}{2} + C \right) \\ &= -\frac{1}{2} + C e^{t^2}\end{aligned}$$

b) Find the specific solution of the given initial-value problem:

$$\frac{dy}{dt} - 2ty = t, \quad y(0) = 1.$$

Solution: Method I: put $y(0)=1$ into the general solution obtained in a), we get

$$\begin{aligned}y(0) &= -\frac{1}{2} + C e^0 = 1 \\ \Rightarrow C &= \frac{3}{2}\end{aligned}$$

$$\text{So } y(t) = -\frac{1}{2} + \frac{3}{2} e^{t^2}$$

Method 2: From a), we know $\frac{d}{dt}[e^{-t^2}y] = e^{-t^2}t$, so

$$\int_0^t \frac{d}{ds}[e^{-s^2}y(s)] ds = \int_0^t e^{-s^2} s ds$$

$$e^{-t^2}y(t) - e^0 y(0) = -\frac{e^{-s^2}}{2} \Big|_{s=0}^{s=t}$$

$$e^{-t^2}y(t) = 1 + \left(-\frac{e^{-t^2}}{2}\right) - \left(-\frac{1}{2}\right)$$

$$y(t) = e^{t^2} \left(-\frac{e^{-t^2}}{2} + \frac{3}{2}\right) = -\frac{1}{2} + \frac{3}{2} e^{t^2}$$